

GNU History Library

Edition 8.2, for History Library Version 8.2.
September 2022

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This document describes the GNU History library (version 8.2, 19 September 2022), a programming tool that provides a consistent user interface for recalling lines of previously typed input.

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1 Using History Interactively

This chapter describes how to use the GNU History Library interactively, from a user's standpoint. It should be considered a user's guide. For information on using the GNU History Library in your own programs, see Section "Programming with GNU History" in *GNU History Library*.

1.1 History Expansion

The History library provides a history expansion feature that is similar to the history expansion provided by `csh`. This section describes the syntax used to manipulate the history information.

History expansions introduce words from the history list into the input stream, making it easy to repeat commands, insert the arguments to a previous command into the current input line, or fix errors in previous commands quickly.

History expansion takes place in two parts. The first is to determine which line from the history list should be used during substitution. The second is to select portions of that line for inclusion into the current one. The line selected from the history is called the *event*, and the portions of that line that are acted upon are called *words*. Various *modifiers* are available to manipulate the selected words. The line is broken into words in the same fashion that Bash does, so that several words surrounded by quotes are considered one word. History expansions are introduced by the appearance of the history expansion character, which is `'!`' by default.

History expansion implements shell-like quoting conventions: a backslash can be used to remove the special handling for the next character; single quotes enclose verbatim sequences of characters, and can be used to inhibit history expansion; and characters enclosed within double quotes may be subject to history expansion, since backslash can escape the history expansion character, but single quotes may not, since they are not treated specially within double quotes.

1.1.1 Event Designators

An event designator is a reference to a command line entry in the history list. Unless the reference is absolute, events are relative to the current position in the history list.

- | | |
|----------------------------------|--|
| ! | Start a history substitution, except when followed by a space, tab, the end of the line, or <code>'=</code> '. |
| !<i>n</i> | Refer to command line <i>n</i> . |
| !<i>-n</i> | Refer to the command <i>n</i> lines back. |
| !! | Refer to the previous command. This is a synonym for <code>'!-1</code> '. |
| !<i>string</i> | Refer to the most recent command preceding the current position in the history list starting with <i>string</i> . |
| !<i>?string</i>[<i>?</i>] | Refer to the most recent command preceding the current position in the history list containing <i>string</i> . The trailing <code>'?</code> ' may be omitted if the <i>string</i> is followed immediately by a newline. If <i>string</i> is missing, the string from the most recent search is used; it is an error if there is no previous search string. |

`^string1^string2^`

Quick Substitution. Repeat the last command, replacing *string1* with *string2*.
Equivalent to `!!:s^string1^string2^`.

`!#` The entire command line typed so far.

1.1.2 Word Designators

Word designators are used to select desired words from the event. A ‘:’ separates the event specification from the word designator. It may be omitted if the word designator begins with a ‘^’, ‘\$’, ‘*’, ‘-’, or ‘%’. Words are numbered from the beginning of the line, with the first word being denoted by 0 (zero). Words are inserted into the current line separated by single spaces.

For example,

`!!` designates the preceding command. When you type this, the preceding command is repeated in toto.

`!!: $` designates the last argument of the preceding command. This may be shortened to `!$`.

`!fi:2` designates the second argument of the most recent command starting with the letters `fi`.

Here are the word designators:

`0 (zero)` The 0th word. For many applications, this is the command word.

`n` The *n*th word.

`^` The first argument; that is, word 1.

`$` The last argument.

`%` The first word matched by the most recent ‘*?string?*’ search, if the search string begins with a character that is part of a word.

`x-y` A range of words; ‘-y’ abbreviates ‘0-y’.

`*` All of the words, except the 0th. This is a synonym for ‘1-\$’. It is not an error to use ‘*’ if there is just one word in the event; the empty string is returned in that case.

`x*` Abbreviates ‘x-\$’

`x-` Abbreviates ‘x-\$’ like ‘x*’, but omits the last word. If ‘x’ is missing, it defaults to 0.

If a word designator is supplied without an event specification, the previous command is used as the event.

1.1.3 Modifiers

After the optional word designator, you can add a sequence of one or more of the following modifiers, each preceded by a ‘:’. These modify, or edit, the word or words selected from the history event.

`h` Remove a trailing pathname component, leaving only the head.

t	Remove all leading pathname components, leaving the tail.
r	Remove a trailing suffix of the form ' <i>.suffix</i> ', leaving the basename.
e	Remove all but the trailing suffix.
p	Print the new command but do not execute it.
s/ <i>old/new</i> /	Substitute <i>new</i> for the first occurrence of <i>old</i> in the event line. Any character may be used as the delimiter in place of '/'. The delimiter may be quoted in <i>old</i> and <i>new</i> with a single backslash. If '&' appears in <i>new</i> , it is replaced by <i>old</i> . A single backslash will quote the '&'. If <i>old</i> is null, it is set to the last <i>old</i> substituted, or, if no previous history substitutions took place, the last <i>string</i> in a <i>!?string[?]</i> search. If <i>new</i> is null, each matching <i>old</i> is deleted. The final delimiter is optional if it is the last character on the input line.
&	Repeat the previous substitution.
g	
a	Cause changes to be applied over the entire event line. Used in conjunction with 's', as in <i>gs/old/new/</i> , or with '&'.
G	Apply the following 's' or '&' modifier once to each word in the event.

2 Programming with GNU History

This chapter describes how to interface programs that you write with the GNU History Library. It should be considered a technical guide. For information on the interactive use of GNU History, see Chapter 1 [Using History Interactively], page 1.

2.1 Introduction to History

Many programs read input from the user a line at a time. The GNU History library is able to keep track of those lines, associate arbitrary data with each line, and utilize information from previous lines in composing new ones.

A programmer using the History library has available functions for remembering lines on a history list, associating arbitrary data with a line, removing lines from the list, searching through the list for a line containing an arbitrary text string, and referencing any line in the list directly. In addition, a history *expansion* function is available which provides for a consistent user interface across different programs.

The user using programs written with the History library has the benefit of a consistent user interface with a set of well-known commands for manipulating the text of previous lines and using that text in new commands. The basic history manipulation commands are similar to the history substitution provided by `csH`.

The programmer can also use the Readline library, which includes some history manipulation by default, and has the added advantage of command line editing.

Before declaring any functions using any functionality the History library provides in other code, an application writer should include the file `<readline/history.h>` in any file that uses the History library's features. It supplies extern declarations for all of the library's public functions and variables, and declares all of the public data structures.

2.2 History Storage

The history list is an array of history entries. A history entry is declared as follows:

```
typedef void *histdata_t;

typedef struct _hist_entry {
    char *line;
    char *timestamp;
    histdata_t data;
} HIST_ENTRY;
```

The history list itself might therefore be declared as

```
HIST_ENTRY **the_history_list;
```

The state of the History library is encapsulated into a single structure:

```
/*
 * A structure used to pass around the current state of the history.
 */
typedef struct _hist_state {
    HIST_ENTRY **entries; /* Pointer to the entries themselves. */
```

```

    int offset;           /* The location pointer within this array. */
    int length;           /* Number of elements within this array. */
    int size;             /* Number of slots allocated to this array. */
    int flags;
} HISTORY_STATE;

```

If the flags member includes HS_STIFLED, the history has been stifled.

2.3 History Functions

This section describes the calling sequence for the various functions exported by the GNU History library.

2.3.1 Initializing History and State Management

This section describes functions used to initialize and manage the state of the History library when you want to use the history functions in your program.

void using_history (void) [Function]
 Begin a session in which the history functions might be used. This initializes the interactive variables.

HISTORY_STATE * history_get_history_state (void) [Function]
 Return a structure describing the current state of the input history.

void history_set_history_state (HISTORY_STATE *state) [Function]
 Set the state of the history list according to *state*.

2.3.2 History List Management

These functions manage individual entries on the history list, or set parameters managing the list itself.

void add_history (const char *string) [Function]
 Place *string* at the end of the history list. The associated data field (if any) is set to NULL. If the maximum number of history entries has been set using **stifle_history()**, and the new number of history entries would exceed that maximum, the oldest history entry is removed.

void add_history_time (const char *string) [Function]
 Change the time stamp associated with the most recent history entry to *string*.

HIST_ENTRY * remove_history (int which) [Function]
 Remove history entry at offset *which* from the history. The removed element is returned so you can free the line, data, and containing structure.

histdata_t free_history_entry (HIST_ENTRY *histent) [Function]
 Free the history entry *histent* and any history library private data associated with it. Returns the application-specific data so the caller can dispose of it.

HIST_ENTRY * replace_history_entry (*int which*, *const char *line*, [Function]
histdata_t data)

Make the history entry at offset *which* have *line* and *data*. This returns the old entry so the caller can dispose of any application-specific data. In the case of an invalid *which*, a NULL pointer is returned.

void clear_history (*void*) [Function]

Clear the history list by deleting all the entries.

void stifle_history (*int max*) [Function]

Stifle the history list, remembering only the last *max* entries. The history list will contain only *max* entries at a time.

int unstifle_history (*void*) [Function]

Stop stifling the history. This returns the previously-set maximum number of history entries (as set by `stifle_history()`). The value is positive if the history was stifled, negative if it wasn't.

int history_is_stifled (*void*) [Function]

Returns non-zero if the history is stifled, zero if it is not.

2.3.3 Information About the History List

These functions return information about the entire history list or individual list entries.

HIST_ENTRY ** history_list (*void*) [Function]

Return a NULL terminated array of HIST_ENTRY * which is the current input history. Element 0 of this list is the beginning of time. If there is no history, return NULL.

int where_history (*void*) [Function]

Returns the offset of the current history element.

HIST_ENTRY * current_history (*void*) [Function]

Return the history entry at the current position, as determined by `where_history()`. If there is no entry there, return a NULL pointer.

HIST_ENTRY * history_get (*int offset*) [Function]

Return the history entry at position *offset*. The range of valid values of *offset* starts at `history_base` and ends at `history_length - 1` (see Section 2.4 [History Variables], page 9). If there is no entry there, or if *offset* is outside the valid range, return a NULL pointer.

time_t history_get_time (*HIST_ENTRY *entry*) [Function]

Return the time stamp associated with the history entry *entry*. If the timestamp is missing or invalid, return 0.

int history_total_bytes (*void*) [Function]

Return the number of bytes that the primary history entries are using. This function returns the sum of the lengths of all the lines in the history.

2.3.4 Moving Around the History List

These functions allow the current index into the history list to be set or changed.

- int history_set_pos** (*int pos*) [Function]
 Set the current history offset to *pos*, an absolute index into the list. Returns 1 on success, 0 if *pos* is less than zero or greater than the number of history entries.
- HIST_ENTRY * previous_history** (*void*) [Function]
 Back up the current history offset to the previous history entry, and return a pointer to that entry. If there is no previous entry, return a NULL pointer.
- HIST_ENTRY * next_history** (*void*) [Function]
 If the current history offset refers to a valid history entry, increment the current history offset. If the possibly-incremented history offset refers to a valid history entry, return a pointer to that entry; otherwise, return a BNULL pointer.

2.3.5 Searching the History List

These functions allow searching of the history list for entries containing a specific string. Searching may be performed both forward and backward from the current history position. The search may be *anchored*, meaning that the string must match at the beginning of the history entry.

- int history_search** (*const char *string, int direction*) [Function]
 Search the history for *string*, starting at the current history offset. If *direction* is less than 0, then the search is through previous entries, otherwise through subsequent entries. If *string* is found, then the current history index is set to that history entry, and the value returned is the offset in the line of the entry where *string* was found. Otherwise, nothing is changed, and a -1 is returned.
- int history_search_prefix** (*const char *string, int direction*) [Function]
 Search the history for *string*, starting at the current history offset. The search is anchored: matching lines must begin with *string*. If *direction* is less than 0, then the search is through previous entries, otherwise through subsequent entries. If *string* is found, then the current history index is set to that entry, and the return value is 0. Otherwise, nothing is changed, and a -1 is returned.
- int history_search_pos** (*const char *string, int direction, int pos*) [Function]
 Search for *string* in the history list, starting at *pos*, an absolute index into the list. If *direction* is negative, the search proceeds backward from *pos*, otherwise forward. Returns the absolute index of the history element where *string* was found, or -1 otherwise.

2.3.6 Managing the History File

The History library can read the history from and write it to a file. This section documents the functions for managing a history file.

- int read_history** (*const char *filename*) [Function]
 Add the contents of *filename* to the history list, a line at a time. If *filename* is NULL, then read from `~/.history`. Returns 0 if successful, or `errno` if not.

int read_history_range (*const char *filename, int from, int to*) [Function]
 Read a range of lines from *filename*, adding them to the history list. Start reading at line *from* and end at *to*. If *from* is zero, start at the beginning. If *to* is less than *from*, then read until the end of the file. If *filename* is NULL, then read from *~/.history*. Returns 0 if successful, or **errno** if not.

int write_history (*const char *filename*) [Function]
 Write the current history to *filename*, overwriting *filename* if necessary. If *filename* is NULL, then write the history list to *~/.history*. Returns 0 on success, or **errno** on a read or write error.

int append_history (*int nelements, const char *filename*) [Function]
 Append the last *nelements* of the history list to *filename*. If *filename* is NULL, then append to *~/.history*. Returns 0 on success, or **errno** on a read or write error.

int history_truncate_file (*const char *filename, int nlines*) [Function]
 Truncate the history file *filename*, leaving only the last *nlines* lines. If *filename* is NULL, then *~/.history* is truncated. Returns 0 on success, or **errno** on failure.

2.3.7 History Expansion

These functions implement history expansion.

int history_expand (*char *string, char **output*) [Function]
 Expand *string*, placing the result into *output*, a pointer to a string (see Section 1.1 [History Interaction], page 1). Returns:

- 0 If no expansions took place (or, if the only change in the text was the removal of escape characters preceding the history expansion character);
- 1 if expansions did take place;
- 1 if there was an error in expansion;
- 2 if the returned line should be displayed, but not executed, as with the **:p** modifier (see Section 1.1.3 [Modifiers], page 2).

If an error occurred in expansion, then *output* contains a descriptive error message.

char * get_history_event (*const char *string, int *cindex, int qchar*) [Function]
 Returns the text of the history event beginning at *string + *cindex*. **cindex* is modified to point to after the event specifier. At function entry, *cindex* points to the index into *string* where the history event specification begins. *qchar* is a character that is allowed to end the event specification in addition to the “normal” terminating characters.

char ** history_tokenize (*const char *string*) [Function]
 Return an array of tokens parsed out of *string*, much as the shell might. The tokens are split on the characters in the *history_word_delimiters* variable, and shell quoting conventions are obeyed as described below.

char * history_arg_extract (*int first, int last, const char *string*) [Function]
 Extract a string segment consisting of the *first* through *last* arguments present in *string*. Arguments are split using *history_tokenize*.

2.4 History Variables

This section describes the externally-visible variables exported by the GNU History Library.

- int history_base** [Variable]
The logical offset of the first entry in the history list.
- int history_length** [Variable]
The number of entries currently stored in the history list.
- int history_max_entries** [Variable]
The maximum number of history entries. This must be changed using `stifle_history()`.
- int history_write_timestamps** [Variable]
If non-zero, timestamps are written to the history file, so they can be preserved between sessions. The default value is 0, meaning that timestamps are not saved.
The current timestamp format uses the value of *history_comment_char* to delimit timestamp entries in the history file. If that variable does not have a value (the default), timestamps will not be written.
- char history_expansion_char** [Variable]
The character that introduces a history event. The default is '!'. Setting this to 0 inhibits history expansion.
- char history_subst_char** [Variable]
The character that invokes word substitution if found at the start of a line. The default is '^'.
- char history_comment_char** [Variable]
During tokenization, if this character is seen as the first character of a word, then it and all subsequent characters up to a newline are ignored, suppressing history expansion for the remainder of the line. This is disabled by default.
- char * history_word_delimiters** [Variable]
The characters that separate tokens for `history_tokenize()`. The default value is "`\t\n()<>;&|`".
- char * history_search_delimiter_chars** [Variable]
The list of additional characters which can delimit a history search string, in addition to space, TAB, ':' and '?' in the case of a substring search. The default is empty.
- char * history_no_expand_chars** [Variable]
The list of characters which inhibit history expansion if found immediately following *history_expansion_char*. The default is space, tab, newline, carriage return, and '='.
- int history_quotes_inhibit_expansion** [Variable]
If non-zero, the history expansion code implements shell-like quoting: single-quoted words are not scanned for the history expansion character or the history comment character, and double-quoted words may have history expansion performed, since single quotes are not special within double quotes. The default value is 0.

int history_quoting_state [Variable]

An application may set this variable to indicate that the current line being expanded is subject to existing quoting. If set to `'`, the history expansion function will assume that the line is single-quoted and inhibit expansion until it reads an unquoted closing single quote; if set to `"`, history expansion will assume the line is double quoted until it reads an unquoted closing double quote. If set to zero, the default, the history expansion function will assume the line is not quoted and treat quote characters within the line as described above. This is only effective if *history_quotes_inhibit_expansion* is set.

rl_linebuf_func_t * history_inhibit_expansion_function [Variable]

This should be set to the address of a function that takes two arguments: a `char *` (*string*) and an `int` index into that string (*i*). It should return a non-zero value if the history expansion starting at *string[i]* should not be performed; zero if the expansion should be done. It is intended for use by applications like Bash that use the history expansion character for additional purposes. By default, this variable is set to NULL.

2.5 History Programming Example

The following program demonstrates simple use of the GNU History Library.

```
#include <stdio.h>
#include <readline/history.h>

main (argc, argv)
    int argc;
    char **argv;
{
    char line[1024], *t;
    int len, done = 0;

    line[0] = 0;

    using_history ();
    while (!done)
    {
        printf ("history$ ");
        fflush (stdout);
        t = fgets (line, sizeof (line) - 1, stdin);
        if (t && *t)
        {
            len = strlen (t);
            if (t[len - 1] == '\n')
                t[len - 1] = '\0';
        }

        if (!t)
            strcpy (line, "quit");

        if (line[0])
        {
            char *expansion;
            int result;

            result = history_expand (line, &expansion);
```

```

        if (result)
            fprintf (stderr, "%s\n", expansion);

        if (result < 0 || result == 2)
        {
            free (expansion);
            continue;
        }

        add_history (expansion);
        strncpy (line, expansion, sizeof (line) - 1);
        free (expansion);
    }

    if (strcmp (line, "quit") == 0)
        done = 1;
    else if (strcmp (line, "save") == 0)
        write_history ("history_file");
    else if (strcmp (line, "read") == 0)
        read_history ("history_file");
    else if (strcmp (line, "list") == 0)
    {
        register HIST_ENTRY **the_list;
        register int i;

        the_list = history_list ();
        if (the_list)
            for (i = 0; the_list[i]; i++)
                printf ("%d: %s\n", i + history_base, the_list[i]->line);
    }
    else if (strncmp (line, "delete", 6) == 0)
    {
        int which;
        if ((sscanf (line + 6, "%d", &which)) == 1)
        {
            HIST_ENTRY *entry = remove_history (which);
            if (!entry)
                fprintf (stderr, "No such entry %d\n", which);
            else
            {
                free (entry->line);
                free (entry);
            }
        }
        else
        {
            fprintf (stderr, "non-numeric arg given to 'delete'\n");
        }
    }
}
}
}

```

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Version 1.3, 3 November 2008

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